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Tutorial: 15

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Restaurant POS System

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Overview

This project aims to develop a **Command-Line Interface (CLI) based Restaurant Point of Sale (POS)** System that manages restaurant operations such as order processing, billing, inventory tracking, and employee management.

The system will be implemented using **Python** and integrated with **PostgreSQL** for persistent data storage. All interactions will occur through a terminal interface, ensuring focus on backend logic, data structures, and database integration.

Objectives

1. Build a fully functional **CLI-based POS system**
2. Implement database operations using **PostgreSQL**
3. Apply programming concepts (**OOP, loops, conditionals, functions**)
4. Simulate **real-world restaurant** workflows
5. Ensure efficient and structured data handling

Key Features

Menu Management

- **Add, update, delete** menu items via CLI commands
- View **categorized** menu
- **Search** items by name or category

Customer Management

- **Add** new customers
- **View** customer details and order history

Billing System

- **Create** orders by selecting menu items
- Automatically **calculate** total cost, tax, and discounts
- **Generate** and display text-based receipts

Inventory Management

- **Track** stock levels of items/ingredients
- **Update** stock of items/ingredients

Table & Order Management

- **Assign** orders to table numbers
- **Track** order status (active/completed)
- **Handle** multiple tables simultaneously

Employee Management

- **Add/manage** employees
- **Assign** roles (waiter, chef, manager)
- **View** employee records

Technologies Used

- **Language:** Python
- **Database:** PostgreSQL
- **Library:** psycopg2 (for database connection)
- **Interface:** Command-Line Interface (CLI)

FlowChart

